Q P Code: 25900516

Mahatma Gandhi University, Kottayam
MGU-UGP (HONOURS)— THIRD SEMESTER EXAMINATION
MG3VACMAT200 - Mastering Problem Solving through Vedic Mathematics
2024 Admission Onwards

Date of Examination: $24-10-2025$ | Time duration: 1 hour 15 Minutes
Paper Code: 25900516 | Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall learning outcome attainability. Each question must be solved using the relevant Vedic sutras with proper steps.

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A

Answer any five questions. Each question carries 2 marks.

1. Using *Ekadhikena Purvena* method, find the sum of $95, 37, 87, 99$ [(**S**)][CO2]

   - $\overline{9}\ 5$
   - $3\ \overline{7}$
   - $\overline{8}\ 7$
   - $\overline{9}\ \overline{9}$
   - 3 1 8 [2 marks]

2. Write the importance of zero in the complement of divisor [(**K**)][CO1]

   - The number of digits to be placed in the remainder column should be equal to the number of zeros in the base. (Any sensible attempt can be awarded marks.) [2 marks]

3. Multiply 2487 by 9999 [(**S**)][CO2]

   - LHS=$2487-1$=2486
   - RHS=[9 9 9 10] − [2 4 8 7] = 7513 [1 mark]
   - Hence, $2487 \times 9999 = 2486\ 7513$ [1 mark]

4. Why is the *Beejank* Method useful in cube root calculations up to 7 digits. [(**U**)] [CO1]

   - Many a time, while extracting the cube root of a number having more than 7 digits, you will get two results for M (Middle term) and will be little confused in deciding which one is the suitable value for M and the Beejank method will be there to help you out. (Any sensible attempt can be given marks) [2 marks]

5. If A:B = 1:2, B:C = 3:4, C:D = 6:9 and D:E = 12:16, find A:B:C:D:E [(**An**)][CO5]

   - 3 : 6 : 8 : 12 : 16 [2 marks]

6. A man buys a toy for Rs 25 and sells it for Rs 30. Find his gain percent [(**A**)] [CO6]

   - Gain $= 30 - 25 = 5$
   - 5 is 20% of 25 [2 marks]

7. Solve $\frac{2}{x+1} + \frac{3}{x+2} = 0$, and find the value of x [(**A**)][CO7]

   - $x = \frac{-mb-na}{m+n}$ [1 mark]
   - $i.e., x = \frac{-4-3}{3+2}$
   - $x = -\frac{7}{5}$ [1 mark]

8. Solve: $3x + 2y = 5$ and $6x + 8y = 10$ [(**U**)][CO8]

   - $x = \frac{2\times10-5\times8}{2\times6-3\times8} = \frac{5}{3}$ [1 mark]
   - Thus, $y = 0$ [1 mark]

Part B

Answer any four questions. Each question carries 5 marks.

9. Multiply 9604 by 111 [(**S**)][CO2]

   - 00 9604 00
   - 009|096|960|604|040|400 [2 marks]
   - 9|15|15|10|4|4 [2 marks]
   - Hence, $9604 \times 111 = 1066044$ [1 mark]

10. Using *Nikhilam Navatascaramam* method, subtract 45993 from 87453 [(**U**)][CO1]

    - 8 7 4 5 3−
    - 4 5 9 9 3
    - ———
    - 4 2 5 4 0 [3 marks]
    - ———
    - 4 1 4 6 0 [2 marks]

11. Apply Duplex Method to find the square of 45612 [(**A**)][CO5]

    - The groups are 4, 45, 456, 4561, 45612, 5612, 612, 12, and 2 [1 mark]
    - Duplex of 4 = 16, 45 = 40, 456 = 73, 4561 = 68, 45612 = 62, 5612 =32, 612 = 25, 12 =4 and 2 = 4 [2 marks]
    - Square is 208,0454,544 [2 marks]

12. At what rate % compound interest does a sum of money become 9-fold in 2 ys? [(**A**)] [CO5]

    - Any reasonable attempt [5 marks]

13. Find the product of $2x^2 + 3y$ and $2x^2 + 5y$ using *Urdhva Tiryagbhyam* [(**A**)] [CO7]

- $2 \times 2|2 \times 5 + 3 \times 2|3 \times 5$ [3 marks]
- $4x^4 + 16x^2y + 15y^2$ [2 marks]

14. Solve $\frac{3}{x+1} + \frac{4}{x+2} = \frac{7}{x+3}$ using *Paravartya* Method [(**An**)][CO7]

- $\frac{3}{x+1} + \frac{4}{x+2} = \frac{7}{x+3} = \frac{3}{x+3} + \frac{4}{x+3}$ [3 marks]
- $\frac{3}{x+1} - \frac{3}{x+3} = \frac{4}{x+3} - \frac{4}{x+2}$
- $\frac{6}{x+1} = -\frac{4}{x+2}$
- $i.e., x = \frac{-4-12}{10} = -\frac{8}{5}$ [2 marks]

OR

- $x = \frac{-mb-na}{m+n}$ [3 marks]
- $i.e., x = \frac{-12-4}{10} = -\frac{8}{5}$ [2 marks]

Part C

Answer any two questions. Each question carries 10 marks.

15. Subtract 122245 from 513323 using digit separator method. [(**S**)][CO2]

5	1	3	3	2	3	
1	2	2	2	4	5	
4	$(-1)^9$	1	1	$(-2)^8$	$(-2)^8$	[6 marks]
	-1			-1	-1	[3 marks]
3	9	1	0	7	8	[1 mark]

16. Determine whether 62932451 is divisible by 139 using the Vedic method of osculators [(**U**)][CO1]

- *Ekadhika* of 139 is 14 [4 marks]
- $(1 \times 14) + 5 \equiv 19 \mod 139$ [2 marks]
- $(19 \times 14) + 4 \equiv 131 \mod 139$ [.5 mark]
- $(131 \times 14) + 2 \equiv 29 \mod 139$ [.5 mark]
- $(29 \times 14) + 3 \equiv 131 \mod 139$ [.5 mark]
- $(131 \times 14) + 9 \equiv 36 \mod 139$ [.5 mark]
- $(36 \times 14) + 2 \equiv 89 \mod 139$ [.5 mark]
- $(89 \times 14) + 6 \equiv 1 \mod 139$ [.5 mark]
- Thus, 62932451 is not divisible by 139 [1 mark]

17. The population of a town is 10,000. It increases by 10% during the first year. During the second year, it decreases by 20% and increased by30% during the third year. What is the population after 3 years? [(**A**)][CO3]

- End of the first year increase (10% of 10,000) = 1,000. The total is 11,000 [5 marks]
- End of the second year decrease (20% of 11,000) = 2,200. The total is 8,800 [3 marks]
- End of the third year increase (30% of 8,800) = 2,640. The total is 11,440 [2 marks]

OR

- $\frac{P(100+x)(100-y)(100+z)}{100\times100\times100}$ [6 marks]
- $\frac{10{,}00\times(100+10)\times(100-20)\times(100+30)}{100\times100\times100} = 11,440$ [4 marks]

18. Multiply the following using *Urdhva Tiryagbhyam*: [(**U**)][CO1]
(a) $(2x^2 - 3x + 4) \times (x^2 + 5x - 6)$

- $2|\ -1|\ 4$
- $1|\ 5\ |-6$
- $x^4|x^3|x^2|x^1|x^0$
- $2\times1|2\times5-3\times1|2\times-6+-3\times5+4\times1|-3\times-6+4\times5|4\times-6$ [3 marks]
- $2x^4+7x^3-23x^2+38x-24$ [2 marks]

(b) $(4x^2 - 1) \times (x^2 - 3x + 7)$

- $4|\ 0|\ -1$
- $1|\ -3\ |7$
- $x^4|x^3|x^2|x^1|x^0$
- $4\times1|4\times-3|4\times7+-1\times1|-1\times-3|-1\times7$ [3 marks]
- $4x^4-12x^3+27x^2+3x-7$ [2 marks]
